

WEEK-2

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Step 1 — Create Database

```
1 • CREATE DATABASE IF NOT EXISTS bookflow_db;
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	21:00:55	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected

Step 2 — Select Database

```
1 • CREATE DATABASE IF NOT EXISTS bookflow_db;
2 • USE bookflow_db;
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	21:00:55	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
✓ 2	21:01:22	USE bookflow_db	0 row(s) affected

Step 3— Create Books Table (DDL + Constraints)

```
1 • CREATE DATABASE IF NOT EXISTS bookflow1_db;
2 • USE bookflow1_db;
3 • CREATE TABLE Books(
4     book_id INT PRIMARY KEY,
5     title VARCHAR(100) NOT NULL,
6     isbn VARCHAR(20) UNIQUE,
7     published_year INT CHECK(published_year<2027)
8 );
```

#	Time	Action
✓ 17	17:03:51	USE bookflow1_db
✓ 18	17:05:03	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT ...

Step 4- Display Books

```
9 • SELECT * FROM Books;
```

Result Grid				
Filter Rows:				
	book_id	title	isbn	published_year
*	NULL	NULL	NULL	NULL

Step 5— Insert books and display

```
10 • INSERT INTO books(title,isbn,published_year)VALUES
11   ('The Alchemist','9374679545265',1988),
12   ('Clean Code','859956356965',2008),
13   ('Atomic habits','3897503565036',2015);
14 • DESCRIBE Books;
```

Result Grid						
Filter Rows:						
Export: Wrap Cell Contents:						
	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	
	title	varchar(100)	NO		NULL	
	isbn	varchar(20)	YES	UNI	NULL	
	published_year	int	YES		NULL	

Step 6—Members table (create, insert, display)

```
15 • CREATE TABLE Members(
16   member_id INT PRIMARY KEY,
17   full_name VARCHAR(100),
18   email VARCHAR(150) UNIQUE
19 );
20 • SELECT * FROM Members;
```

Result Grid			
Filter Rows:			
	member_id	full_name	email
*	NULL	NULL	NULL

```
21 • INSERT INTO Members(member_id,full_name,email)VALUES
22   ('101','likhitha','likhitha@gmail.com'),
23   ('102','kavya','kavya@gmail.com'),
24   ('103','akshitha','akshitha@gmail.com');
25 • SELECT * FROM Members;
```

Result Grid			
Filter Rows:			
Edit: Export:			
	member_id	full_name	email
▶	101	likhitha	likhitha@gmail.com
	102	kavya	kavya@gmail.com
	103	akshitha	akshitha@gmail.com
*	NULL	NULL	NULL

Step 7— Create Loans table(FOREIGN KEYS)

```
26 CREATE TABLE Loans(  
27     loan_id INT PRIMARY KEY,  
28     member_id INT,  
29     book_id INT,  
30     Loan_date DATE,  
31     FOREIGN KEY (member_id) REFERENCES Member_id,  
32     FOREIGN KEY (book_id) REFERENCES Book_id,  
33 );  
34 SELECT * FROM Loans;
```

Result Grid

Filter Rows:

Edit:

	loan_id	member_id	book_id	Loan_date
*	NULL	NULL	NULL	NULL

Step 8— Insert data into loans and display

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),  
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),  
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),  
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');  
SELECT * FROM Loans;
```

	loan_id	member_id	book_id	loan_date
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01
*	NULL	NULL	NULL	NULL

Step 7— GROUP BY Query(books per year)

```
SELECT published_year, COUNT(book_id) AS Total_Books  
FROM Books  
GROUP BY published_year  
ORDER BY published_year;
```

	published_year	Total_Books
▶	1925	1
	1949	1
	1960	1

Step 8 — Donation via TRANSACTION(atomicity)

```
CREATE TABLE Donation_History (
  donation_id INT PRIMARY KEY,
  book_id INT,
  donor_name VARCHAR(100),
  donation_date DATE,
  FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
START TRANSACTION;
INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);
INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());
COMMIT;
```

```
SELECT * FROM Books;
```

	book_id	title	isbn	published_year
▶	1	The Great Gatsby	9780743273565	1925
	2	To Kill a Mockingbird	9780061120084	1960
	3	1984	9780451524935	1949
	4	Animal Farm	9780451526342	1945
*	NULL	NULL	NULL	NULL

SELECT * FROM Donation_History;

	donation_id	book_id	donor_name	donation_date
▶	1	4	Raj Kumar	2026-08-10
	NULL	NULL	NULL	NULL

Step 9 – Index on ISBN + FAST SEARCH

```
CREATE INDEX idx_books_isbn ON Books(isbn);
SELECT * FROM Books WHERE isbn = '9780451524935';
```

	book_id	title	isbn	published_year
▶	3	1984	9780451524935	1949
*	NULL	NULL	NULL	NULL